

# Lab 8: Cellular Genetics — Mitosis and Meiosis

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BIOL-1

Name: \_\_\_\_\_ Date: \_\_\_\_\_

## Learning Objectives

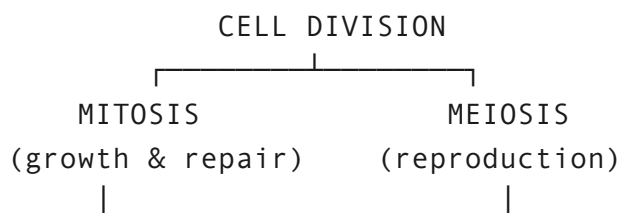
By the end of this lab, you will be able to:

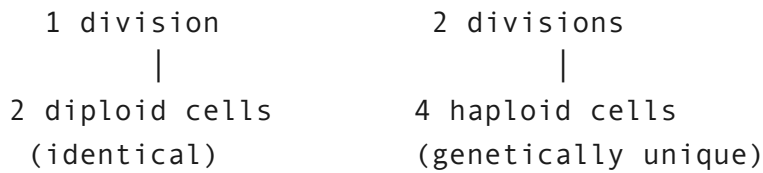
1. **Define** key vocabulary: diploid, haploid, homologous chromosomes, sister chromatids, allele, gamete.
2. **Explain** what happens to chromosomes during interphase (S phase) before division begins.
3. **Model mitosis** step by step using paper chromosomes, tracing each allele through every phase.
4. **Model meiosis I and II** using paper chromosomes, distinguishing what separates at each stage.
5. **Explain** the critical difference between Anaphase I (homologs separate) and Anaphase II (sister chromatids separate).
6. **Compare and contrast** mitosis and meiosis by purpose, number of divisions, chromosome number, genetic outcome, and other features.
7. **Explain** why meiosis produces genetically unique gametes while mitosis produces identical daughter cells.

## Overview

Every cell in your body came from a single fertilized egg that divided over and over again.

That process — **cell division** — is how you grow, repair wounds, and replace worn-out cells. But not all cell division works the same way:





In this lab, you will use **paper chromosomes** — RED for maternal, BLUE for paternal — to physically walk through both processes and see exactly where each allele ends up.

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## Key Terms

| Term                          | Definition                                                                             |
|-------------------------------|----------------------------------------------------------------------------------------|
| <b>Diploid (2n)</b>           | A cell with two full sets of chromosomes (one from each parent)                        |
| <b>Haploid (n)</b>            | A cell with one set of chromosomes                                                     |
| <b>Homologous chromosomes</b> | Matching chromosome pairs — one maternal (red), one paternal (blue)                    |
| <b>Sister chromatids</b>      | Two identical copies of one chromosome, joined at the centromere after DNA replication |
| <b>Allele</b>                 | A version of a gene (e.g., <b>A</b> = dominant, <b>a</b> = recessive)                  |
| <b>Centromere</b>             | The region where sister chromatids are joined together                                 |
| <b>Gamete</b>                 | A sex cell (egg or sperm), produced by meiosis                                         |

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## Materials

- **Chromosome Cutouts** (last page of this lab)
- Chromosome Pair 1: Gene A (alleles **A** and **a**)
- Chromosome Pair 2: Gene B (alleles **B** and **b**)
- Blank paper for staging (draw large circles to represent cells)

**No scissors needed.** Carefully tear along the dashed borders of each chromosome card. Slow, steady tears work best — go corner to corner.

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## Part 1: Chromosome Basics

Before you tear out the chromosomes, answer these questions to check your vocabulary.

1. Match each term to its definition:

| Term                   | Definition           |
|------------------------|----------------------|
| Chromatin              | <input type="text"/> |
| Chromosome             | <input type="text"/> |
| Sister chromatids      | <input type="text"/> |
| Homologous chromosomes | <input type="text"/> |
| Diploid (2n)           | <input type="text"/> |
| Haploid (n)            | <input type="text"/> |

2. What is the difference between sister chromatids and homologous chromosomes?

## Setting Up Your Cell

We are modeling a simple organism with **2 pairs of chromosomes** ( $2n = 4$ ). The genotype of your starting cell is **AaBb**.

Chromosome Pair 1 (Gene A):

|          |          |
|----------|----------|
| Maternal | Paternal |
| A        | a        |
| [RED]    | [BLUE]   |

Chromosome Pair 2 (Gene B):

|          |          |
|----------|----------|
| Maternal | Paternal |
| B        | b        |



Starting cell genotype: AaBb (diploid,  $2n = 4$ )

## Procedure

1. Tear out the last page and cut out the 4 **Starting Cell** chromosomes.
2. Draw a large circle on blank paper to represent your cell.
3. Place all 4 chromosomes inside the circle. This is your **diploid parent cell ( $2n = 4$ )**.

### 3. Record the genotype of your starting cell:

### 4. How many chromosomes are in this cell? Is it diploid or haploid?

## Part 2: Mitosis — Making Identical Copies

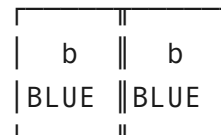
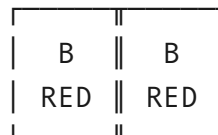
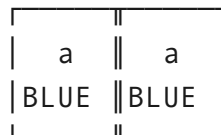
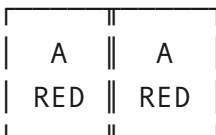
*Mitosis is used for growth, repair, and replacing old cells. It produces two genetically identical diploid cells.*

### Step 1: S Phase (DNA Replication)

Before mitosis begins, each chromosome is copied.

1. Cut out the 4 **S-Phase Duplicate** chromosomes from the cutouts page.
2. **Tape** each duplicate to its matching original at the center — these become **sister chromatids**.
3. You now have 4 duplicated chromosomes (8 chromatids total) inside one cell.

After S Phase — each chromosome is now an X-shaped pair of sister chromatids. On your desk it should look like this:



## Step 2: Prophase

1. Chromosomes condense into visible X-shapes (sister chromatids joined at centromere).
2. Arrange all 4 duplicated chromosomes loosely inside your cell circle.
3. **Note:** In mitosis, homologous chromosomes do **not** pair up.

## Step 3: Metaphase

1. **Line up** all 4 duplicated chromosomes single-file along the middle of the cell (the metaphase plate).

## Step 4: Anaphase

1. **Separate the sister chromatids** — pull each taped pair apart.
2. Move one chromatid from each pair to one side of the cell, the other to the opposite side.
3. Each side should now have 4 chromatids (one A, one a, one B, one b).

## Step 5: Telophase & Cytokinesis

1. **Draw a line** down the middle to separate the two groups into two daughter cells.
2. Each daughter cell now has 4 chromosomes (AaBb,  $2n = 4$ ) — identical to the parent.

## Mitosis Results — Allele Tracking

| # | Cell | Gene A Allele(s) | Gene B Allele(s) | Full Genotype | Diploid or Haploid? | # Chromosomes |
|---|------|------------------|------------------|---------------|---------------------|---------------|
| 1 |      |                  |                  |               |                     |               |
| 2 |      |                  |                  |               |                     |               |
| 3 |      |                  |                  |               |                     |               |

**5. Are the two daughter cells genetically identical to each other and to the parent cell?**

6. Did the chromosome number change? Explain what happened.

|  |
|--|
|  |
|--|

7. Why is mitosis described as producing "clones" of the parent cell?

|  |
|--|
|  |
|--|

## Part 3: Meiosis I — Separating Homologous Chromosomes

*Reset your cell: gather all chromosomes back together. Re-tape sister chromatids if needed. You should again have 4 duplicated chromosomes inside one diploid cell ( $AaBb$ ,  $2n = 4$ ).*

### Step 1: Prophase I

Chromosomes condense. **Homologous chromosomes pair up** side by side — this is called **synapsis**.

1. Bring the A chromosome (red) next to the a chromosome (blue) — this is one **homologous pair**.
2. Bring the B chromosome (red) next to the b chromosome (blue) — this is the other **homologous pair**.
3. Each pair is called a **tetrad** (4 chromatids in close contact).

*(Note: In real cells, crossing over can happen here — homologs exchange segments of DNA. We are keeping it simple and skipping that step today.)*

### Step 2: Metaphase I — Homolog Pairs Line Up

1. **Line up the two tetrads** (homologous pairs) along the middle of the cell as pairs, not as individual chromosomes.

### Step 3: Anaphase I

1. **Separate the homologous pairs** — pull each tetrad apart into its two individual (still-duplicated) chromosomes.
2. Move one complete chromosome from each pair to each side.
3. **Important:** Sister chromatids stay together! Only homologs separate here.

## Step 4: Telophase I & Cytokinesis I

1. **Divide the cell** into two new cells (draw two new circles).
2. Each new cell has **2 duplicated chromosomes** (sister chromatids still joined).
3. Each cell is now **haploid ( $n = 2$ )**, but chromosomes are still duplicated.

## Meiosis I Results — Allele Tracking

| # | Cell | Chromosome Pair 1 Allele | Chromosome Pair 2 Allele | Ploidy | # Chromosomes (each still duplicated) |
|---|------|--------------------------|--------------------------|--------|---------------------------------------|
| 1 |      |                          |                          |        |                                       |
| 2 |      |                          |                          |        |                                       |
| 3 |      |                          |                          |        |                                       |

**8. What separated during Anaphase I — homologous chromosomes or sister chromatids?**

**9. After Meiosis I, are the cells diploid or haploid? How many chromosomes does each cell have?**

## Part 4: Meiosis II — Separating Sister Chromatids

*Perform the following steps for EACH of the two cells from Meiosis I.*

### Step 1: Metaphase II

1. In each cell, line up the duplicated chromosomes individually along the middle of the cell.

## Step 2: Anaphase II

1. **Separate the sister chromatids** — pull them apart at the tape.
2. Move one chromatid to each side.

## Step 3: Telophase II & Cytokinesis II

1. **Divide each cell** into two new cells.
2. You should now have **4 total cells** (gametes), each with **2 single chromosomes ( $n = 2$ )**.

## Final Meiosis Results — All 4 Gametes

| # | Cell | Gene A Allele | Gene B Allele | Genotype | Haploid? | # Chromosomes |
|---|------|---------------|---------------|----------|----------|---------------|
| 1 |      |               |               |          |          |               |
| 2 |      |               |               |          |          |               |
| 3 |      |               |               |          |          |               |
| 4 |      |               |               |          |          |               |

**10. What separated during Anaphase II — homologous chromosomes or sister chromatids?**

**11. How many gametes did you produce? Are they diploid or haploid?**

**12. Are any of the four gametes genetically identical to each other? Explain.**



**13. Why is it important that gametes are haploid (n) instead of diploid (2n)? What would happen at fertilization if both egg and sperm were diploid?**

## Part 5: Mitosis vs. Meiosis — Side-by-Side Comparison

*Review your results and complete the comparison table.*

### Mitosis vs. Meiosis Comparison

| # | Feature | Mitosis | Meiosis |
|---|---------|---------|---------|
| 1 |         |         |         |
| 2 |         |         |         |
| 3 |         |         |         |
| 4 |         |         |         |
| 5 |         |         |         |
| 6 |         |         |         |
| 7 |         |         |         |

**14. In mitosis, the parent cell (AaBb) produced daughter cells with genotype(s):**

**15. In meiosis, the parent cell (AaBb) produced gametes with genotype(s):**

**16. Which process produced more genetic variety? Why?**

**17. Summarize the key difference between mitosis and meiosis in your own words (2–3 sentences):**

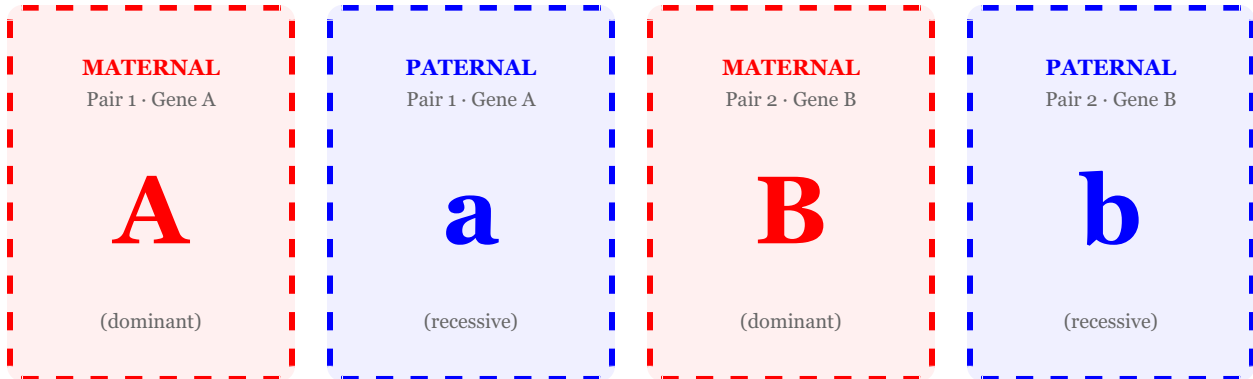
**18. A human skin cell is diploid ( $2n = 46$ ). If it undergoes mitosis, how many chromosomes will each daughter cell have? If a human reproductive cell ( $2n = 46$ ) undergoes meiosis, how many chromosomes will each gamete have?**

# Chromosome Cutouts — Tear Out This Page

Cut along the dashed lines. **RED = Maternal** | **BLUE = Paternal**

## Starting Cell Chromosomes

Use these 4 chromosomes to build your starting diploid cell ( $AaBb$ ,  $2n = 4$ ).



## S-Phase Duplicates (Sister Chromatids)

Keep these aside until S-Phase. Tape each duplicate to its matching chromosome above at the center to form sister chromatids.

